

OJASVA MISHRA

AI/Robotics Engineer · Founder · Researcher · UN Advocate
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EDUCATION

University of Florida — B.S. Computer Engineering + M.S. Finance (4-year combined pathway) Fall 2026
Full merit scholarship

Downingtown STEM Academy – IB Diploma Candidate

Class of 2026

GPA: 4.45 / 4.0 Weighted · Pathway: Robotics I–III

IB HL: Math AA, Physics, Economics, Design Technology · IB SL: German, Literature, Theory of Knowledge

Athletics: 4-Year Varsity Tennis — led team drills; reached Districts for first time in school history

RESEARCH & TECHNICAL EXPERIENCE

Carnegie Mellon University Robotics Institute

Jul 2025 – Present

Research Intern — Prof. Min Xu & Dr. Xiaolong Wu

- ▶ First-author on two peer-reviewed papers in robotic joint control and safety filtering:
- ▶ **Paper 1:** "[Constraint-Aware Discrete-Time PID Gain Optimization for Robotic Joint Control Under Actuator Saturation](#)" — arXiv preprint, under review at IEEE Transactions on Robotics (TRO)
- ▶ **Paper 2:** "[SCIG++: Risk-Limiting Conformal Safety Filtering for Attack-Resilient Robotic Joint Control Under Saturation](#)" — under review at IEEE IROS; algorithms demonstrated on Franka Emika Panda robot ([supplementary video](#))
- ▶ Multi-modal SLAM path planning research with PID forest structure implementation; additional recommendation letter from Dr. Xiaolong Wu

godrift.ai — AI Platform Engineer (Founding Team)

2025 – Present

Drift – Antler VC-backed AI copilot for robotics simulation (ROS2, Gazebo, Ignition) · godrift.ai

- ▶ Joined as founding AI engineer after connecting with founders at NexHacks; work from ojasva@godrift.ai
- ▶ Drift enables natural-language setup, debugging, and execution of robot simulations — think "Claude Code for robotics"
- ▶ Reached #5 Product of the Day on Product Hunt at public launch; platform now used by thousands of engineers
- ▶ Founders became meaningful mentors; directly shaped my approach to AI infrastructure and developer tooling

EV Bots Inc — AI Robotics Engineer Intern

Apr 2025 – Present

EV Bots – Autonomous EV charging robots · Multiple provisional patents

- ▶ Shipped first fully working autonomous EV charging robot prototype after iterating across mechanical, electronics, and controls — [Latest Prototype](#)
- ▶ Designed CAD for autonomous drone charging dock system targeting swarm drone delivery fleets

PteroLabs — Engineering Intern

Jul 2025 – Jan 2026

PteroLabs – Mission-grade UAV simulation

- ▶ Developed mission-grade UAV simulator in Unreal Engine with drone racing capabilities and AI/ML for mission planning & object detection
- ▶ Performance-benchmarked simulation software now used by Penn for active research

WORK EXPERIENCE

iPipeline — Software Development Intern

Summer 2025

- ▶ Built automation testing frameworks and AI-driven agents; deployed internal tools at scale

Kumon Learning Center — Center Assistant

2022 – 2023

- ▶ Tutored students ages 4–16 in math and English; graded coursework and managed student data

Paid Tennis Coach & Volunteer Manager

2022 – 2025

- ▶ Coached students ages 8–15; led drills, assisted varsity coach, and managed the high school team

FOUNDED VENTURES

GlassBoxAI — Co-Founder & CEO

2024 – Present

- ▶ Won Startup in a Weekend Philly hackathon (sole high schooler among 200+ university competitors — UPenn, Drexel, MIT) with GlassBoxAI
- ▶ Interview integrity platform using real-time eye-tracking, facial landmarks, and voice pattern analysis to detect AI-assisted answers (ChatGPT, Cluely, etc.) live over Zoom or Google Meet

- Combines multimodal behavioral AI, computer vision, audio analysis, and language models; garnered VC/angel investor interest — raised \$20K pre-seed

Kinetix — Founder

2025 – Present

- AI-powered physical therapy and athletic training coach using real-time computer vision + joint-angle biomechanics + voice feedback + Meta Quest integration
- Demo: youtube.com/watch?v=-k4Ks9YATTI

PreventaAI — Founder

2023 – Present

- Proactive AI healthcare iOS app (SwiftUI + Firebase) — medication tracker, AI health assistant, visual scanner, quizzes
- Published on App Store; user-reported case where the app helped catch severe tuberculosis earlier than standard screening
- App Store: [PreventaAI](#)

Eyesight — Founder

2025 – Present

- Accessibility app for visually impaired users: LiDAR, CoreML, offline inference, obstacle detection, and real-time navigation/surroundings narration
- Beta: [TestFlight](#)

QuickCitePro — Founder

2025 – Present

- Chrome extension solving a real research pain point; adopted by professors and lab teammates
- Chrome Web Store: [QuickCitePro](#)

FusionBots — Co-Founder & CEO (Non-Profit)

Dec 2022 – Present

- Founded robotics non-profit — 200+ students, \$6K+ revenue; created affordable robotics kits and published free 120-page robotics & programming textbook — [SSRN paper](#)
- Team of 18 — fusionbots.org

LEADERSHIP & ACTIVITIES

TSA — National Competitor

Ongoing

- Competed in 8 events at nationals, each with placement:
- **SAFMS:** Built AI/IoT modelling web app for agroforestry carbon trading using React + blockchain — [Live App](#)
- **Drone Challenge:** Engineered and tuned a Betaflight FPV drone with custom CAD & wiring — [Portfolio](#)
- **Engineering Design:** Designed nitrogen-capture domes using zeolites to reduce agricultural emissions — [Portfolio](#)
- **Systems Control:** Coded autonomous control systems onsite in Python and C++ under timed constraints
- **Robotics:** Built and programmed an autonomous/drivable object-manipulating robot
- VR Visualization: Used Unity 6 to develop a fully playable escape room game on Meta Quest — [Portfolio](#)
- Video Game Design: Built tower defense game in Unity with C# and custom Blender/PixelArt assets — [Portfolio](#)

VEX Robotics — Captain & Lead Programmer

2019 – Present

- Led CAD, C++/Python autonomous routines, PID tuning, and modular subsystems; multi-award competitive seasons

CTO — Youth Publications & Socioeconomic Forum (UN MGCY NGO)

2024 – Present

- Led global AI literacy programs spanning 15+ countries; presented at UN conferences (HLPF, UNGA, etc.); 500+ global members — ypsforum.org

American Rocketry Club — President & Founder

2023 – Present

- Founded 60+ member club; leads CAD payloads, propulsion analysis, and launches

BattleBots Club — Co-President

2022 – Present

- Designed and built combat robots; drivetrain, armor, and strategy

Stocks & Investments Club — CIO

2022 – Present

- Built AI Robo-Trader analyzing SPY for ODTE options — 300%+ returns over 2 years;
- Grew to 100+ members with AI stock-tracking tools and workshops

Chess Competitor & Coach

2014 – Present

- 1800+ ELO; 8K YouTube subscribers; 30+ OTB/online USCF-rated tournament wins; coached 50+ students

National Honor Society — Member

2023 – Present

- Selected for academic excellence, leadership, and community service; organized volunteer initiatives and peer mentorship

NexHacks

2025

- One of 3 high school students accepted among 1,500+ hackers — trajectory-defining networking event that led to Drift opportunity

Varsity Tennis — 4 Years

2022 – Present

- Led team drills and strategy; reached Districts for first time in school history

SKILLS

Robotics: VEX, Arduino, UAVs, PID, SLAM, Odometry, ROS2, Gazebo

CAD: AutoCAD, OnShape, Fusion360, RockSim, Revit

Programming: Python, C++, Swift, JavaScript, C#, HTML/CSS

Frameworks/Tools: React, Unity, Firebase, Git, Blender, Unreal Engine, 3D Printing

AI/ML: Computer Vision, LLMs, Multimodal Models, CoreML, Reinforcement Learning, OpenClaw

Other: Public Speaking, Technical Writing, Financial Literacy, Violin, Piano, German

HONORS & AWARDS

- ▶ Startup in a Weekend Philly — Winner (sole high schooler; 200+ university competitors from UPenn, Drexel, MIT)
- ▶ TSA Nationals — 8 events with National/State level placement
- ▶ NexHacks — 1 of 3 high schoolers accepted (1,500+ hackers)
- ▶ CalcuSolve Math Competition — 1st Place Team, 2nd Place Individual
- ▶ CollegeBoard National Recognition · NM Commended Scholar · NHS Member
- ▶ DECA & HOSA State Qualifier
- ▶ VEX Robotics — Multi-award competitive seasons
- ▶ 30+ USCF-rated chess tournament top-5 finishes